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DOCUMENT NAME:

FORA Interface Control Document

D/N: Rev:

FILE NAME: TICD_FORA_BPMeter Update: 2015/07/28

Revision History

Date. Done by. The update. New rev.

1 2011/08/23 Carol Primary Edition. Ver1.0

2 2011/10/14 Carol Modification. Ver1.1

3 2013/06/03 Carol Command 0x54 modification. Ver1. 2

4 2014/01/09 Carol Modify command 0x25 Ver1.3

5 2015/04/16 Carol Modify command 0x25 Ver1.4

6 2015/07/28 Carol Note for different Bluetooth ver sions. Ver1.5

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1. General

1.1. Overview

The document describes the commands which will be u sed to communicate with blood pressure meter.

Before retrieving data in the device, please make s ure you have already established the proper connection channel between the device and the host side. For example, if you use USB cable, be sure that the USB driver has been properly installe d; or if you use Bluetooth communication, be sure that the connection has been successfully esta blished.

Generally speaking, the device will be the slave si de waiting for the request from the host which is usually the computer or the hub/gateway.

Note for different Bluetooth versions:

For Bluetooth v2 meter: Please connect to meter by SPP (Serial Port Profile). Meter always stands by for both pairing and connection, the PIN for par ing is 111111.

For Bluetooth v3 meter: Please connect to meter by SPP (Serial Port Profile). Meter must be paired before any connection, please find the instr uction in meter manual for the pairing process. The meter supports SSP (Simple Secure Pairing), as a result PIN is not required. If host does not support SSP, please use 0000 as PIN if required.

For Bluetooth v4 meter: Please connect to meter by following GATT service:

UUID Base: 1212-efde-1523-785feabcd123

Service UUID: 0x1523

Characteristic: 0x1524 (write/notify)

1.2. Document Overview

For the first time user on the device integration, please read the 2.1 frame structure. If you are familiar with serial port communication, you can re view the command list table in section 2.2 directly. The following sub sections are all about the details for each command.

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1.3. Definitions, Acronyms, and Abbreviations

GW Gateway / Hub / Computer side

MD Medical Device, ie. blood glucose meter, blood pressure meter or scales, etc.

CMD Command . Usually it's a one byte alphanumeric symbol.

ACK Acknowledgement. Usually it's a one byte alphanumeric symbol.

NA Not available

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2. Interface specification

Block Diagram

The communication interface is based on RS232 serial port communication :

2.1. Frame Structure

Byte 1 2 3 ~ n-2 n-1 n

Name Start CMD Index, Address, or Data Stop Check
Sum

Format HEX

GW MD 51 CMD A3 [1..n-1]

GW MD 51 ACK A5 [1..n-1]

ie. Check sum means the summation from byte 1 to byte n-1 with 8-bit length

Example : 8-byte frame

Byte 1 2 3 4 5 6 7 8

Name Start CMD Data_0 Data_1 Data_2 Data_3 Stop ChkSum

Format HEX

GW MD 51 CMD Data_0 Data_1 Data_2 Data_3 A3 [1..7]

GW MD 51 ACK Data_0 Data_1 Data_2 Data_3 A5 [1..7]

ie. Check sum means the summation from byte 1 to byte 7 with 8-bit length Baud rate 19200

Data bits 8

Parity No

Start bit 1

Stop bit 1 Medical Device /

Sensor

(MD) Gateway / Hub /

Computer

(GW) TX

RX

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2.2. Command List

Message name CMD/ACK Direction Length
(Bytes) Note

Read device clock time 0x23 GW MD 8

0x23 GW MD 8

Read device model 0x24 GW MD 8

0x24 GW MD 8

Read storage data, part 1 (time) 0x25 GW MD 8

0x25 GW MD 8

Read storage data, part 2(result) 0x26 GW MD 8

0x26 GW MD 8

Read device serial number, part 1 0x27 GW MD 8

0x27 GW MD 8

Read device serial number, part 2 0x28 GW MD 8

0x28 GW MD 8

Read storage number of data 0x2B GW MD 8

0x2B GW MD 8

Write device clock time 0x33 GW MD 8

0x33 GW MD 8

Turn off the device 0x50 GW MD 8

0x50 GW MD 8

Clear/Delete all memory 0x52 GW MD 8

0x52 GW MD 8

2.2.1. [0x23] Read device clock time (4 bytes)

Message Name: Read device clock time (4 bytes)

Message ID: 0x23

Message Description: Read device date and time (to minute)

Src / Des : GW / MD

Length: Request: 8 bytes / Response: 8 bytes

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Format

Message name Direction CMD Data_0 Data_1 Data_2 Data_3

Read System Clock time GW MD 0x23 0x0 0x0 0x0 0x0

Response from device. GW MD 0x23 Day + Month + Year Minute Hour

Field Description

Field Size
 (Bytes) Value Description.
 CMD 1 0x23 GW commands MD to response device clock time
 DATA 4 0x0 NA

Field Size
 (Bytes) Value Description.
 ACK 1 0x23 MD response device clock time
 DATA 2 Data_1 + Data_0 : Day+Month+Year (word) (see Table A)
 DATA 2 Data_2 : Minute, Data_3 : Hour (see Table B)

Table A: Day + Month + Year

Table B: Minute and Hour

MSB	LSB		
Data_1	Data_0		
Year (7-bit) Month (4-bit) Day (5-bit)			
6	5	4	3
2	1	0	3
2	1	0	4
3	2	1	0
MSB	LSB		
Data_3	Data_2		
NA Hour (5-bit)		NA Minute (6-bit)	
	4	3	2
	1	0	5
			4
			3
			2
			1
			0

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 2.2.2. [0x24] Read device model
 Message Name: Read device model
 Message ID: 0x24
 Message Description: Read device model / project code
 Src / Des : GW / MD
 Length: Request: 8 bytes / Response: 8 bytes

Format
 Message name Direction CMD Data_0 Data_1 Data_2 Data_3
 Read System Clock time GW MD 0x24 0x0 0x0 0x0 0x0
 Response from device. GW MD 0x2 4 Project Code

Field Description
 # Field Size
 (Bytes) Value Description.
 CMD 1 0x24 GW commands MD to response device model
 DATA 4 0x0

Field Size
 (Bytes) Value Description.
 ACK 1 0x24 MD response project code
 DATA 2 Data_1 + Data_0 : Device model in four digits (Data_1 is the higher byte)
 DATA 2 Data_3, Data_2 : undefined

2.2.3. [0x25] Read the storage data with index, part 1(time)
Message Name: Read the storage data with index, part 1(time)
Message ID: 0x25
Message Description: Read data (part 1: measure date and time)
Src / Des: GW / MD
Length: Request: 8 bytes / Response: 8 bytes

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Format

Message name Direction CMD Data_0 Data_1 Data_2 Data_3
Read System Clock time GW MD 0x25 Index 0x0 User
Response from device. GW MD 0x25 M_Date M_Time_Type

Field Description

Field Size

(Bytes) Value Description.

CMD 1 0x25 GW commands MD to response measure data (part 1)

Index 2 0x00 is the latest reading in the device

DATA 1 0x00 NA

DATA 1 Meter is single user User=0x00;

Meter is multiple user:

User=0x1 means User1;

User=0x2 means User2;

User=0x3 means User3;

User=0x4 means User4

Field Size

(Bytes) Value

Description.

ACK 1 0x25 MD response measure date and time

M_Date 2 The measure date (see Table A in 2.2.1 [0x23])

M_Time_Type 1 Data_2 : Minute + IHB

IHB for Td-3128: (see Table C)

IHB =0 : Normal Heart Beats

IHB =1 : IHB

IHB for Td-3140: (see Table D)

IHB =00 : Normal Heart Beats

IHB =01 : Tachycardia(>110) or Bradycardia(<50)

IHB =10 : Varied Heart Rate (20%)

IHB=11 : Atrial Fibrillation (AF)

Data_3 : Hour + Avg+ User (see Table C, D)

AVG=0 : Single measurement reading

AVG=1 : Average measurement reading

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User=00,01,10,11 : user 1~ user 4

Table C: M_Time_Type for Td-3128

Table D: M_Time_Type for Td-3140

2.2.4. [0x26] Read the storage data with index, part 2(result)
Message Name: Read the storage data with index, part 2(result)
Message ID: 0x26
Message Description: Read data (part 2: measure result)
Src / Des : GW / MD
Length: Request: 8 bytes / Response: 8 bytes

Format
Message name Direction CMD Data_0 Data_1 Data_2 Data_3
Read System Clock time GW MD 0x26 Index 0x0 User
Response from device. GW MD 0x26 Systolic MAP Diastolic Pulse

Field Description
Field Size
(Bytes) Value Description.
CMD 1 0x26 GW commands MD to response measure data (part 1)
Index 2 0x00 is the latest reading in the device MSB LSB
Data_3 Data_2
Avg User Hour (5-bit) NA IHB Minute (6-bit)
0 1 0 4 3 2 1 0 0 0 5 4 3 2 1 0
MSB LSB
Data_3 Data_2
Avg User Hour (5-bit) IHB Minute (6-bit)
0 1 0 4 3 2 1 0 1 0 5 4 3 2 1 0

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DATA 1 0x00 NA
DATA 1 User=0x1 means User1;
User=0x2 means User2;
User=0x3 means User3;
User=0x4 means User4

Field Size
(Bytes) Value Description.
ACK 1 0x26 MD response measure date and time
Systolic 1 The measure result (see Table E below)

The unit is mmHg (Kpa = mmHg / 7.5)

Mean

pressure 1 The measure result (see Table E below)

The unit is mmHg (Kpa = mmHg / 7.5)

Diastolic 1 The measure result (see Table F below)

The unit is mmHg (Kpa = mmHg / 7.5)

Pulse 1 The measure result (see Table F below)

The unit is pul/min

Table E: Systolic and Mean pressure

Table F: Parameters

MSB LSB

Data_1 Data_0

Mean Pressure (8-bit) (Unit=mmHg) Systolic(8-bit) (Unit=mmHg)

6 5 4 3 2 1 0 3 2 1 0 4 3 2 1 0

MSB LSB

Data_3 Data_2

Pulse (8-bit) (Unit= pul/min) Diastolic (8-bit) (Un it=mmHg)

7 6 5 4 3 2 1 0 7 6 5 4 3 2 1 0

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2.2.5. [0x27] Read device serial number, part 1

Message Name: Read the device serial number, part 1

Message ID: 0x27

Message Description: Read the device serial number, part 1. This command cooperates with 0x28 to complete the serial number.

Src / Des: GW / MD

Length : Request: 8 bytes / Response: 8 bytes

The Serial Number for the device is composed of 16 characters (0~9, A~F) : SN_7 + + SN_0
(example: 32502102 4013102A, the former 4 bytes come from 0x28 and the later come from 0x27)

Format

Message name Direction CMD Data_0 Data_1 Data_2 Data_3

Read device serial

number, part 1 GW MD 0x27 0x0 0x0 0x0 0x0

Response from device GW MD 0x27 SN_0 SN_1 SN_2 SN_3

Field Description

Field Size

(Bytes) Value Description.

CMD 1 0x27 GW commands MD to response the device serial number, part 1

DATA 4 0x0

Field Size

(Bytes) Value Description.

ACK 1 0x27 Read the device serial number, part 1
DATA 4 Data_0 ~ Data_3 : SN_0 ~ SN_3

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2.2.6. [0x28] Read device serial number, part 2

Message Name: Read the device serial number, part 1

Message ID: 0x28

Message Description: Read the device serial number, part 2. This command cooperates with 0x27 to complete the serial number.

Src / Des: GW / MD

Length : Request: 8 bytes / Response: 8 bytes

The Serial Number for the device is composed of 16 characters (0~9, A~F) : SN_7 + + SN_0
(example: 32502102 4013102A, the former 4 bytes come from 0x28 and the later come from 0x27)

Format

Message name Direction CMD Data_0 Data_1 Data_2 Data_3

Read device serial

number, part 2 GW MD 0x28 0x0 0x0 0x0 0x0

Response from device GW MD 0x28 SN_4 SN_5 SN_6 SN_7

Field Description

Field Size

(Bytes) Value Description.

CMD 1 0x28 GW commands MD to response the device serial
number , part 2

DATA 4 0x0

Field Size

(Bytes) Value Description.

ACK 1 0x28 Read the device serial number, part 2

DATA 4 Data_0 ~ Data_3 : SN_4 ~ SN_7

2.2.7. [0x2B] Read storage number of data

Message Name: Read storage number of data

Message ID: 0x2B

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Message Description: Read data counts

Src / Des: GW / MD
Length : Request: 8 bytes / Response: 8 bytes

Format

Message name Direction CMD Data_0 Data_1 Data_2 Data_3
Read Storage Number
of Stored Readings GW MD 0x2B User 0x0 0x0 0x0
Response from device GW MD 0x2B Storage Number 0x0 0x0

Field Description

Field Size

(Bytes) Value Description.

CMD 1 0x2B GW commands MD to response the Number of
Stored Readings

DATA 1 User=0x1 means User1;

User=0x2 means User2;

User=0x3 means User3;

User=0x4 means User4

DATA 3 0x0

Field Size

(Bytes) Value Description.

ACK 1 0x2B MD response the Number of Stored Readings

DATA 2 Data_1+Data_0 = Storage Number (Word)

2.2.8. [0x33] Write System Clock time (4 bytes)

Message Name: Write device clock time (4 bytes)

Message ID: 0x 33

Message Description: Write the device date and time (to minute)

Src / Des: GW / MD

Length: Request: 8 bytes / Response: 8 bytes

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Please refer to 0x23 command section for the format of date and time.

Format

Message name Direction CMD Data_0 Data_1 Data_2 Data_3

Write System Clock

time GW MD 0x33 Day+Month+Year Minute Hour

Response from device. GW MD 0x33 Day+Month+Year Minute Hour

Field Description

Field Size

(Bytes) Value Description.

CMD 1 0x33 GW commands MD to write System Clock time (4
bytes)

DATA 2 Data_1 + Data_0 : Day+Month+Year (word)

DATA 2 Data_2 : Minute, Data_3 : Hour

Field Size
(Bytes) Value Description.
ACK 1 0x33 MD write System Clock time (4 bytes) O.K.
DATA 2 Data_1 + Data_0 : Day+Month+Year (word)
DATA 2 Data_2 : Minute, Data_3 : Hour

2.2.9. [0x50] Turn off the device
Message Name: Turn off the device
Message ID: 0x 50
Message Description: Turn off the device from GW
Src / Des: GW / MD
Length: Request: 8 bytes / Response: 8 bytes

Format
Message name Direction CMD Data_0 Data_1 Data_2 Data_3

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Turn off the device GW MD 0x50 0x0 0x0 0x0 0x0
GW MD 0x50 0x0 0x0 0x0 0x0

Field Description
Field Size
(Bytes) Value Description.
CMD 1 0x50 GW commands MD to turn off the power.
DATA 4 0x0 NA

Field Size
(Bytes) Value Description.
ACK 1 0x 50 MD responses.
DATA 4 0x0 NA

2.2.10. [0x52] Clear/Delete all Memory
Message Name: Clear/Delete all memory
Message ID: 0x52
Message Description: Delete all measurement data in the device
Src / Des: GW / MD
Length: Request: 8 bytes / Response: 8 bytes

Format
Message name Direction CMD Data_0 Data_1 Data_2 Data_3
Clear/Delete all Memory
Storages GW MD 0x52 User 0x0 0x0 0x0
Response from device. GW MD 0x52 User 0x0 0x0 0x0

Field Description
Field Size
(Bytes) Value Description.

CMD 1 0x52 GW commands MD to Clear/Delete all Memory Storages in Device EEPROM

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DATA 1 User=0x1 means User1;
User=0x2 means User2;
User=0x3 means User3;
User=0x4 means User4
DATA 3 0x0 NA

Field Size

(Bytes) Value Description.

ACK 1 0x52 Clear/Delete all Memory Storages in Device EEPROM
O.K.

DATA 1 User
DATA 3 0x0 NA

2.2.11. [0x54] Notification for entering communication mode

Message Name: Notification for entering communication mode

Message ID: 0x54

Message Description: Some models will send 0x54 command to host as a notification to indicate meter is ready for command from host. Host has to deal with this kind of notification in the beginning of connection (like RS-232) and prevent it to be mixed with the expected command response in receiving queue of host.

Src / Des: MD / GW

Length: Request: 8 bytes / Response: NA

Format

Message name Direction CMD Data_0 Data_1 Data_2 Data_3

Notification of entering

communication mode GW MD 0x54 0x0 0x0 0x0 0x0

No need to response from GW side

Field Description

Field Size Value Description.

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(Bytes)

CMD 1 0x54 MD reports to GW that it enters communication mode.

DATA 4 0x0 NA